

VoxShield

Indic-Language Voice-Deepfake Detection under Telephony Conditions — an early-stage research prototype

Current status — early-stage prototype.

We treat the numbers below as evidence to stress-test, not production claims. We would value your view on whether our evaluation methodology is rigorous enough — especially cross-generator generalisation and telephony/codec robustness.

System (implemented).

8 kHz G.711 μ -law → Silero VAD → 3 s window / 1 s hop → **five complementary detectors** (Acoustic DSP · wav2vec2 · XLS-R 300M · DistilHuBERT · LFCC+CQCC) over an 89-dim feature bank at a 16 kHz analysis rate (group-delay phase, prosody, and a >6 kHz band that is empty under 8 kHz telephony) → **learned meta-fusion** → Indic/telephony/general routing → Platt calibration → LOW/MEDIUM/HIGH + reason codes. Decision-support for a fraud workflow; never auto-blocks.

Strongest measured results (with scope).

Result	Value	Evaluation scope
Meta-fusion EER	5.9%	held-out In-the-Wild (cross-dataset)
Best single detector / naive avg.	16.0% / 19.1% EER	same set — fusion beats both
Indic fake recall	42% → 82%	MMS-TTS fakes; English EER unchanged (5.9%)
Genuine-Indic false-positive rate	36.3% → 6.3%	tested Indic conditions; channel-aware
ROC-AUC · calibration	0.983 · ECE 0.044	held-out; calibration distribution

EER is not accuracy; Indic recall is not overall recall. Telephony results are on augmented narrowband, not captured PSTN/SIP; speaker-overlap and generator-disjoint testing are not yet documented.

Where your expertise would most help.

- **Cross-generator generalisation** — our Indic evidence uses one generator (MMS-TTS). Is train-A/B/C → test-unseen-D/E (EER, AUC, TPR@FPR, per-generator) the right benchmark? [connects to your ASVspoof-generalisation survey]
- **Telephony/codec protocol** — clean→codec, a codec→noise→packet-loss cascade, or real captured calls? What baselines? [your replay / instantaneous-frequency feature work]
- **Generator × codec** — could the model learn codec rather than synthesis artifacts? How would you separate them?
- **Speaker disjointness** — should speaker- and generator-disjoint be the principal benchmark?
- **Metrics** — EER / AUC / TPR@1% FPR / t-DCF: which matter most for fraud screening?

What we'd value (low-friction).

Is our evaluation methodology sound? What would you change to establish cross-generator generalisation? Which failure mode should we test first? A detailed 6-page brief (methodology, figures, roadmap) is available if useful.

Thank you for your time. Team DigiSeva · VoxShield · GBT BuildStorm 2026 · devanshgoenka03@gmail.com · GitHub: DG10911/voxshield